Problem 1

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Part 1

Construct the transformation matrices that describe the kinematics of the three center of mass positions of the links.

In [50]:
```
import sympy as sp
from sympy import Matrix, cos, sin, simplify
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# Define Symbols
t  = sp.symbols('t', real=True)
l = sp.symbols('l', real=True)   # Length of Link, assume Links are same length
m = sp.symbols('m', real=True)
k = sp.symbols('k', real=True) # Drag Coefficient
d = sp.symbols('d', real=True)

# Configuration variables as a function of time
x = sp.Function('x')(t)
y = sp.Function('y')(t)
theta = sp.Function('theta')(t)
alpha1 = sp.Function('alpha1')(t)
alpha2 = sp.Function('alpha2')(t)
# Alpha kinematics
omega = sp.symbols("omega",real=True)
A1 = sp.symbols("A1",real=True)
A2 = sp.symbols("A2",real=True)
phi = sp.symbols("phi",real=True)
l = sp.symbols("l", real=True)

q = sp.Matrix([x, y, theta, alpha1, alpha2])
qd = q.diff(t)
qdd = q.diff(t, 2)

# Helper Functions
def cross_matrix(vec):

    return np.array([[0, -vec[2,0], vec[1,0]],
                     [vec[2,0], 0, -vec[0,0]],
                     [-vec[1,0], vec[0,0], 0]])

def uncross_matrix(mat):

    return np.array([[mat[2,1]],
                     [mat[0,2]],
```

```
                    [-mat[0,1]]])

def twist2vec(twist):
    m = Matrix.zeros(6,1)

    m[0:3,0] = uncross_matrix(twist[0:3,0:3])
    m[3:,0] = twist[0:3,3]

    return m

def R_z(theta):
    return np.array([[cos(theta), -sin(theta), 0],
                    [sin(theta),  cos(theta), 0],
                    [0,           0,          1]])

def R_y(theta):
    return np.array([[cos(theta), 0, sin(theta)],
                    [0,          1, 0],
                    [-sin(theta),0, cos(theta)]])

def R_x(theta):
    return np.array([[1, 0, 0],
                    [0, cos(theta), -sin(theta)],
                    [0, sin(theta),  cos(theta)]])

def T(R, p):
    m = Matrix.zeros(4,4)
    m[0:3, 0:3] = R
    m[:3, 3] = p
    m[3, 3] = 1
    return m

# World to body center
T0b = T(R_z(theta), Matrix([[x], [y], [0]]))
T_0b = simplify(T0b)

# Left link COM transform
#T_center_to_left_joint = T(Matrix.eye(3), Matrix([-l/2, 0, 0]))
T_left_joint_rotation  = T(R_z(sp.pi - alpha1), Matrix([-l/2, 0, 0]))
T_left_link_com_offset = T(Matrix.eye(3), Matrix([l/2, 0, 0]))
T_01 = simplify(T_0b @ T_left_joint_rotation @ T_left_link_com_offset)

# Right link COM transform
T_center_to_right_joint = T(Matrix.eye(3), Matrix([l/2, 0, 0]))
T_right_joint_rotation  = T(R_z(alpha2), Matrix([0, 0, 0]))
T_right_link_com_offset = T(Matrix.eye(3), Matrix([l/2, 0, 0]))
T_02 = simplify(T_0b @ T_center_to_right_joint @ T_right_joint_rotation @ T_right_l

print("World to body center transformation matrix:")
T_0b
```

```
World to body center transformation matrix:
```

Out[50]:

$$\begin{bmatrix} \cos(\theta(t)) & -\sin(\theta(t)) & 0 & x(t) \\ \sin(\theta(t)) & \cos(\theta(t)) & 0 & y(t) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

In [51]:

```
print("Left link COM transformation matrix:")
simplify(T_01)
T_01
```

```
Left link COM transformation matrix:
```

Out[51]:

$$\begin{bmatrix} -\cos(\alpha_1(t)-\theta(t)) & -\sin(\alpha_1(t)-\theta(t)) & 0 & -\frac{l\cos(\alpha_1(t)-\theta(t))}{2} - \frac{l\cos(\theta(t))}{2} + x(t) \\ \sin(\alpha_1(t)-\theta(t)) & -\cos(\alpha_1(t)-\theta(t)) & 0 & \frac{l\sin(\alpha_1(t)-\theta(t))}{2} - \frac{l\sin(\theta(t))}{2} + y(t) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

In [8]:

```
print("Right link COM transformation matrix:")
T_02
```

```
Right link COM transformation matrix:
```

Out[8]:

$$\begin{bmatrix} \cos(\alpha_2(t)+\theta(t)) & -\sin(\alpha_2(t)+\theta(t)) & 0 & \frac{l\cos(\alpha_2(t)+\theta(t))}{2} + \frac{l\cos(\theta(t))}{2} + x(t) \\ \sin(\alpha_2(t)+\theta(t)) & \cos(\alpha_2(t)+\theta(t)) & 0 & \frac{l\sin(\alpha_2(t)+\theta(t))}{2} + \frac{l\sin(\theta(t))}{2} + y(t) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Part 2

Generate the body twists for these link center of masses.

In [9]:

```
# Body twist matrices
Vb_hat_1 = simplify(T_01.inv() @ sp.diff(T_01, t))
Vb_hat_b = simplify(T_0b.inv() @ sp.diff(T_0b, t))
Vb_hat_2 = simplify(T_02.inv() @ sp.diff(T_02, t))

# Body twist vectors
Vb_1 = twist2vec(Vb_hat_1)
Vb_b = twist2vec(Vb_hat_b)
Vb_2 = twist2vec(Vb_hat_2)

Vb_b
```

Out[9]:

$$\begin{bmatrix} 0 \\ 0 \\ \frac{d}{dt}\theta(t) \\ \sin(\theta(t))\frac{d}{dt}y(t) + \cos(\theta(t))\frac{d}{dt}x(t) \\ -\sin(\theta(t))\frac{d}{dt}x(t) + \cos(\theta(t))\frac{d}{dt}y(t) \\ 0 \end{bmatrix}$$

In [10]:
```
Vb_1
```

Out[10]:

$$\left[\begin{matrix}0\\0\\-\frac{d}{dt}\alpha_1(t)+\frac{d}{dt}\theta(t)\\\frac{-\frac{l\sin\left(2\alpha_1(t)-\theta(t)\right)\frac{d}{dt}\theta(t)}{2}-\frac{l\sin\left(\theta(t)\right)\frac{d}{dt}\theta(t)}{2}+\sin\left(2\alpha_1(t)-2\theta(t)\right)\frac{d}{dt}y(t)-\cos\left(2\alpha_1(t)-2\theta(t)\right)\frac{d}{dt}x(t)-\frac{d}{dt}x(t)}{2\cos\left(\alpha_1(t)-\theta(t)\right)}\\\frac{l\cos\left(\alpha_1(t)\right)\frac{d}{dt}\theta(t)}{2}-\frac{l\frac{d}{dt}\alpha_1(t)}{2}+\frac{l\frac{d}{dt}\theta(t)}{2}-\sin\left(\alpha_1(t)-\theta(t)\right)\frac{d}{dt}x(t)-\cos\left(\alpha_1(t)-\theta(t)\right)\frac{d}{dt}y(t)\\0\end{matrix}\right]$$

In [11]:
```
Vb_2
```

Out[11]:

$$\left[\begin{matrix}0\\0\\\frac{d}{dt}\alpha_2(t)+\frac{d}{dt}\theta(t)\\\frac{\frac{l\sin\left(2\alpha_2(t)+\theta(t)\right)\frac{d}{dt}\theta(t)}{2}-\frac{l\sin\left(\theta(t)\right)\frac{d}{dt}\theta(t)}{2}+\sin\left(2\alpha_2(t)+2\theta(t)\right)\frac{d}{dt}y(t)+\cos\left(2\alpha_2(t)+2\theta(t)\right)\frac{d}{dt}x(t)+\frac{d}{dt}x(t)}{2\cos\left(\alpha_2(t)+\theta(t)\right)}\\\frac{l\cos\left(\alpha_2(t)\right)\frac{d}{dt}\theta(t)}{2}+\frac{l\frac{d}{dt}\alpha_2(t)}{2}+\frac{l\frac{d}{dt}\theta(t)}{2}-\sin\left(\alpha_2(t)+\theta(t)\right)\frac{d}{dt}x(t)+\cos\left(\alpha_2(t)+\theta(t)\right)\frac{d}{dt}y(t)\\0\end{matrix}\right]$$

Part 3

Generate the body body twist Jacobians that map pose/joint velocities into twists.

In [12]:
```
# Body Jacobians
Jb_1 = simplify(Vb_1.jacobian(qd))
Jb_b = simplify(Vb_b.jacobian(qd))
Jb_2 = simplify(Vb_2.jacobian(qd))

Jb_b
```

Out[12]:

$$\left[\begin{matrix}0&0&0&0&0\\0&0&0&0&0\\0&0&1&0&0\\\cos\left(\theta(t)\right)&\sin\left(\theta(t)\right)&0&0&0\\-\sin\left(\theta(t)\right)&\cos\left(\theta(t)\right)&0&0&0\\0&0&0&0&0\end{matrix}\right]$$

In [13]:
```
Jb_1
```

Out[13]:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ -\frac{\cos(2\alpha_1(t)-2\theta(t))+1}{2\cos(\alpha_1(t)-\theta(t))} & \frac{\sin(2\alpha_1(t)-2\theta(t))}{2\cos(\alpha_1(t)-\theta(t))} & -\frac{l\sin(\alpha_1(t))}{2} & 0 & 0 \\ -\sin(\alpha_1(t)-\theta(t)) & -\cos(\alpha_1(t)-\theta(t)) & \frac{l(\cos(\alpha_1(t))+1)}{2} & -\frac{l}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

In [14]:

```
Jb_2
```

Out[14]:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ \frac{\cos(2\alpha_2(t)+2\theta(t))+1}{2\cos(\alpha_2(t)+\theta(t))} & \frac{\sin(2\alpha_2(t)+2\theta(t))}{2\cos(\alpha_2(t)+\theta(t))} & \frac{l\sin(\alpha_2(t))}{2} & 0 & 0 \\ -\sin(\alpha_2(t)+\theta(t)) & \cos(\alpha_2(t)+\theta(t)) & \frac{l(\cos(\alpha_2(t))+1)}{2} & 0 & \frac{l}{2} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Part 4

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In [66]:

```
k, l = sp.symbols('k l', positive=True, real=True)

def body_wrench(Vb, k, l):
    B = Matrix([
        [0, 0, 0,          0,    0,    0],
        [0, 0, 0,          0,    0,    0],
        [0, 0, 2*k*l**3/3,0,    0,    0],
        [0, 0, 0,          k*l,  0,    0],
        [0, 0, 0,          0,   2*k*l, 0],
        [0, 0, 0,          0,    0,    0]
    ])

    Fb = -B * Vb
    return Fb

# Body wrench vectors
Fb_1 = body_wrench(Vb_1, k, l)
Fb_b = body_wrench(Vb_b, k, l)
Fb_2 = body_wrench(Vb_2, k, l)
```

Part 5

In [67]:

```
# Parameters
omega = sp.symbols("omega",real=True)
A1 = sp.symbols("A1",real=True)
A2 = sp.symbols("A2",real=True)
phi = sp.symbols("phi",real=True)
l = sp.symbols("l",real=True)
```

```
params = [l,k,A1,A2,omega,phi]

def Vb(t,q,q_dt,params):
    return -np.linalg.inv(A11_lam(0,q,q_dt,params)) @ A12_lam(0,q,q_dt,params)

def dynamics(t,state,alpha_fcn,params):
    x,y,theta = state
    l, k, A1, A2, omega, phi = params
    alpha1, alpha2, alpha1_dot, alpha2_dot = alpha_fcn(t, omega, A1, A2, phi)
    q_ = [x,y,theta,alpha1,alpha2]
    q_dt_= [0,0,0,0,0]

    dydt= Vb(t,q_,q_dt_,params_) @ np.matrix(([alpha1_dot],[alpha2_dot]))

    return dydt.flatten().tolist()[0]
```

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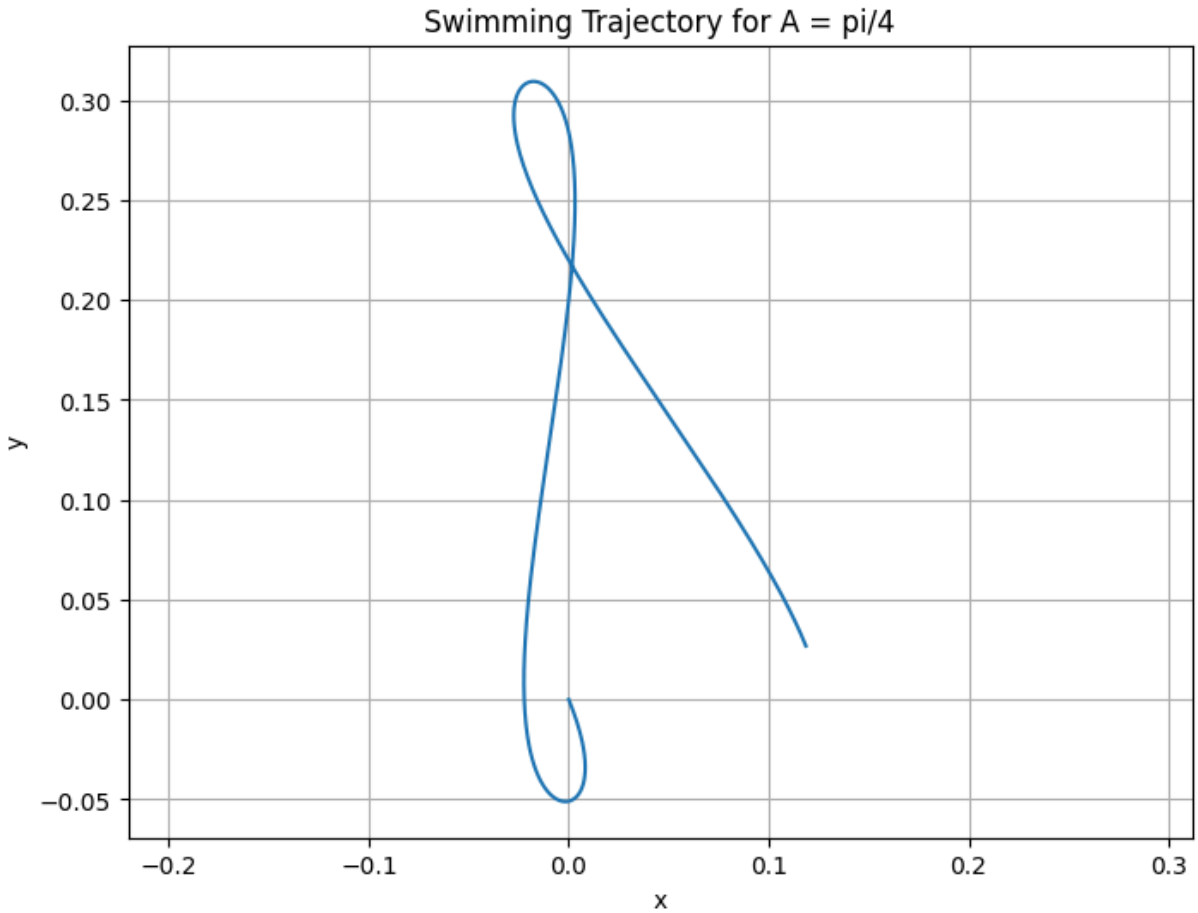
Part 6

In [34]:
```
B = sp.Matrix([
    [0, 0, 0,         0,      0,      0],
    [0, 0, 0,         0,      0,      0],
    [0, 0, 2*k*l**3/3,0,      0,      0],
    [0, 0, 0,         k*l,    0,      0],
    [0, 0, 0,         0,    2*k*l,    0],
    [0, 0, 0,         0,      0,      0]
])

A_mat = (Jb_1.T @ B @ Jb_1) + (Jb_b.T @ B @ Jb_b) + (Jb_2.T @ B @ Jb_2)

A11 = A_mat[:3, :3]
A12 = A_mat[:3, 3:5]

A_tilde = A11.LUsolve(A12)

A11_lam = lambdify([t,q,qd,params],A11)
A12_lam = lambdify([t,q,qd,params],A12)
```

Part 7

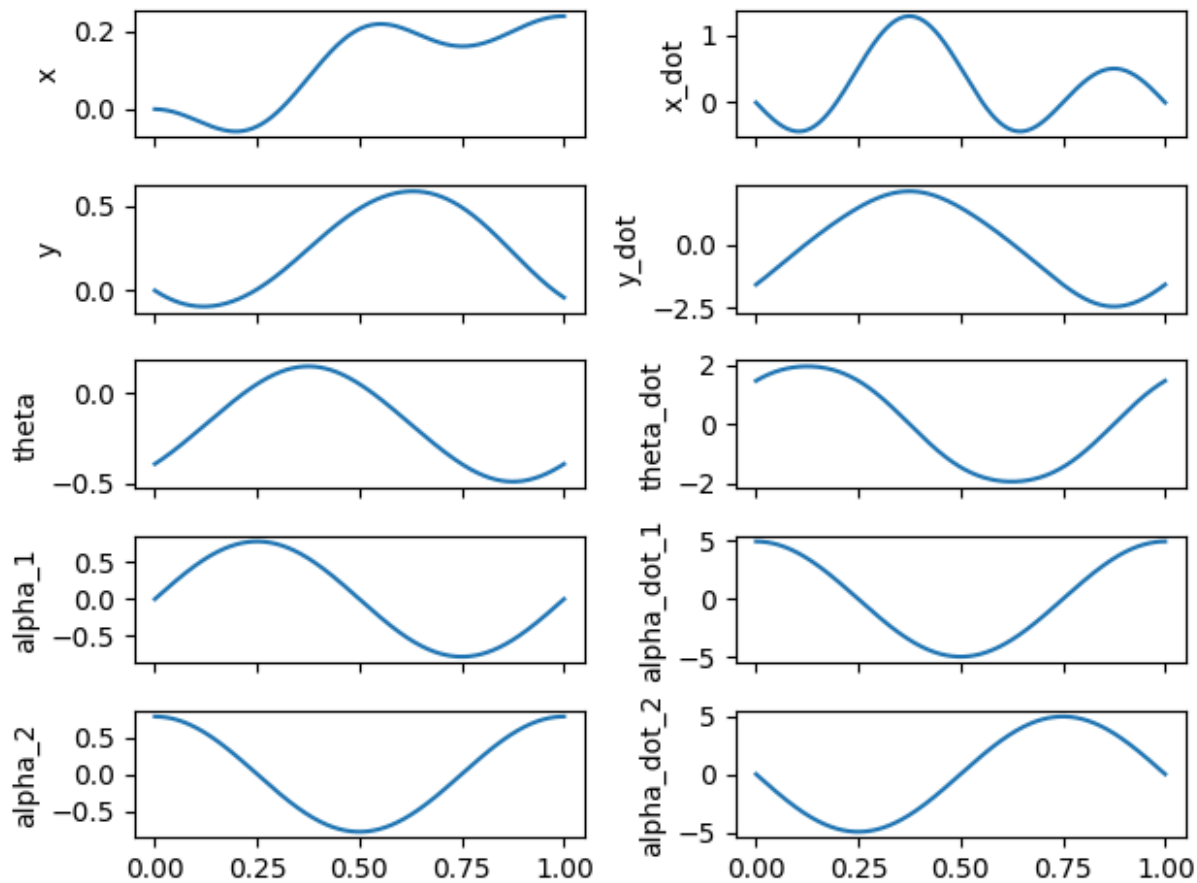
In [35]:
```
from sympy import lambdify
import sympy as sp
from sympy import Matrix, cos, sin, simplify, diff
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# Gait Function
alpha_fcn = lambdify((t, omega, A1, A2, phi),[A1*sin(omega*t),
                                A2*sin(omega*t +phi),
```

```
                                    diff(A1*sin(omega*t),t),
                                    diff(A2*sin(omega*t + phi),t)])
l_=1
k_=1
A_amp = np.pi/4
A1_   = A_amp
A2_   = A_amp
phi_  = np.pi/2
omega_= 2*np.pi
params_ = [l_, k_, A1_, A2_, omega_, phi_]

# Initial conditions [x, y, theta]
y0     = [0, 0, 0]
t_span = (0, 1)
t_eval = np.linspace(0, 1, 500)

# Solve
sol = solve_ivp(
    lambda t, y: dynamics(t, y, alpha_fcn, params_),
    t_span,
    y0,
    t_eval=t_eval,
    rtol=1e-8,
    atol=1e-8
)
# Plot x,y trajectory
plt.figure(figsize=(8, 6))
plt.plot(sol.y[0], sol.y[1])
plt.xlabel('x')
plt.ylabel('y')
plt.title('Swimming Trajectory for A = pi/4')
plt.axis('equal')
plt.grid(True)
plt.show()
```



In [39]:
```
# Initial parameters
l_ = 2
w_ = 1
m_ = 1
k_ = 1
A1_ = np.pi/4
A2_ = A1_
omega_ = 2*np.pi
phi_ = np.pi/2
params_ = [l_, k_, A1_, A2_, omega_, phi_]

init_state = [0, 0, -np.pi/8]
t_end = 1
dt = 0.001
time = np.linspace(0, t_end, int(t_end/dt))

sol = solve_ivp(lambda t, y: dynamics(t, y, alpha_fcn, params_),
                [0, t_end], init_state,
                t_eval=time,
                rtol=1e-8, atol=1e-8)

# Fix: unpack params to call alpha_fcn correctly
l_, k_, A1_, A2_, omega_, phi_ = params_
alphas = alpha_fcn(time, omega_, A1_, A2_, phi_)

q_dots = sol.y * 0
for i, tim in enumerate(time):
```

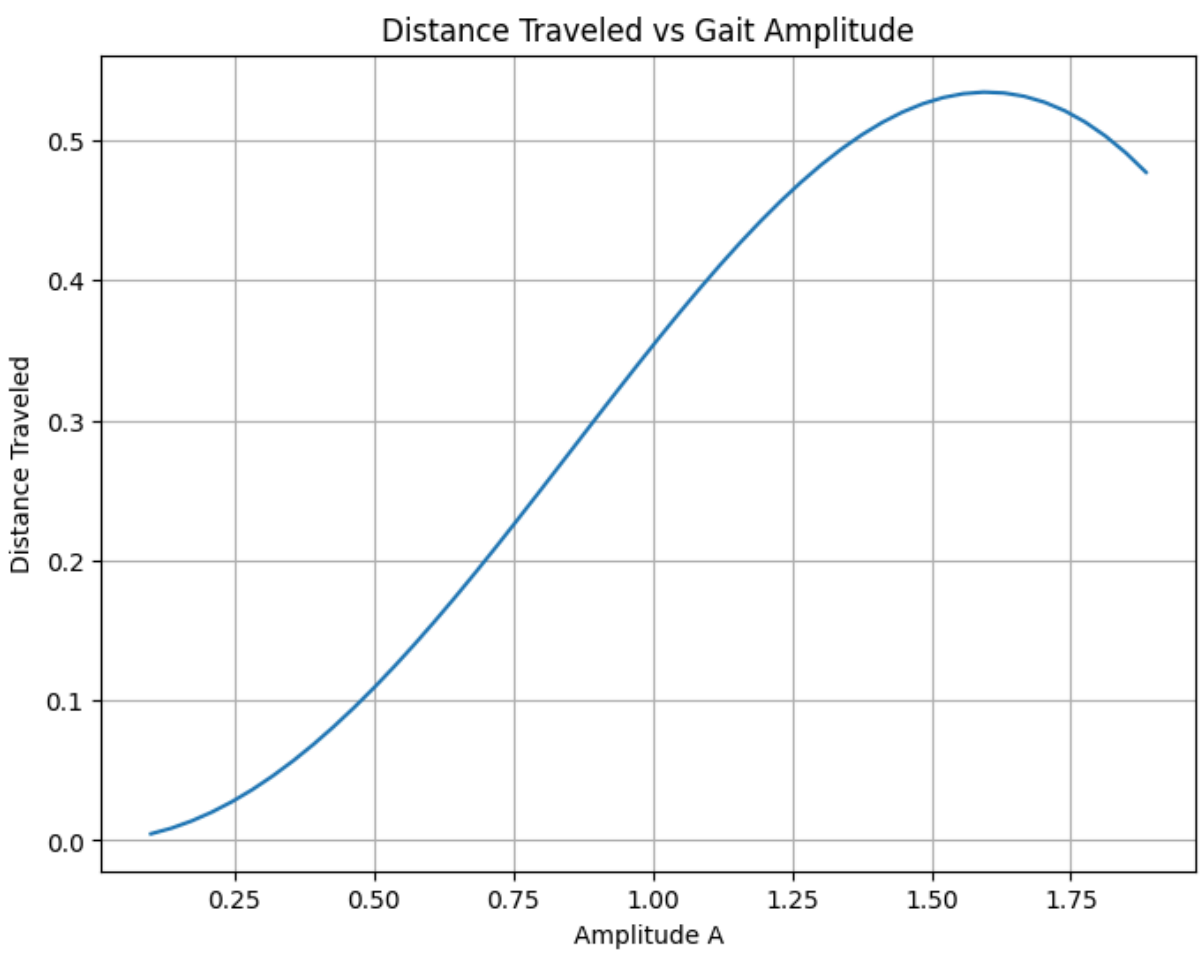
```
    q_temp = dynamics(tim, sol.y[:, i], alpha_fcn, params_)
    q_dots[:, i] = q_temp

x_sol       = sol.y[0, :]
y_sol       = sol.y[1, :]
theta_sol   = sol.y[2, :]
x_dot_sol       = q_dots[0, :]
y_dot_sol       = q_dots[1, :]
theta_dot_sol   = q_dots[2, :]
alpha_0_sol     = alphas[0]
alpha_1_sol     = alphas[1]
alpha_0_dot_sol = alphas[2]
alpha_1_dot_sol = alphas[3]  # Fix: was alphas[2]

fig = plt.figure(2)
fig.clf()
ax = fig.subplots(5, 2, sharex=True)
ax[0][0].plot(time, x_sol);       ax[0][0].set_ylabel('x')
ax[1][0].plot(time, y_sol);       ax[1][0].set_ylabel('y')
ax[2][0].plot(time, theta_sol);   ax[2][0].set_ylabel('theta')
ax[3][0].plot(time, alpha_0_sol); ax[3][0].set_ylabel('alpha_1')
ax[4][0].plot(time, alpha_1_sol); ax[4][0].set_ylabel('alpha_2')
ax[0][1].plot(time, x_dot_sol);       ax[0][1].set_ylabel('x_dot')
ax[1][1].plot(time, y_dot_sol);       ax[1][1].set_ylabel('y_dot')
ax[2][1].plot(time, theta_dot_sol);   ax[2][1].set_ylabel('theta_dot')
ax[3][1].plot(time, alpha_0_dot_sol); ax[3][1].set_ylabel('alpha_dot_1')
ax[4][1].plot(time, alpha_1_dot_sol); ax[4][1].set_ylabel('alpha_dot_2')
plt.tight_layout()
plt.show()
```



Part 8

In [42]:
```
# Part 8 - Sweep Amplitude from 0.1 to 1.2*pi/2
A_vals = np.linspace(0.1, 1.2*np.pi/2, 50)
distances = []

phi_ = np.pi/2

for A1_ in A_vals:

    # Parameters
    A2_ = A1_
    params_ = [l_, k_, A1_, A2_, omega_, phi_]

    # Time setup
    init_state = [0, 0, 0]
    t_end = 1
    dt = 0.01
    time = np.linspace(0, t_end, int(t_end/dt))

    # Solve ODE
    sol = solve_ivp(
        lambda t, y: dynamics(t, y, alpha_fcn, params_),
        (0, t_end),
        init_state,
        t_eval=time,
        rtol=1e-8,
```

```
        atol=1e-8
    )

    # velocities
    q_dots = np.zeros((3, len(time)))

    for i, ti in enumerate(time):
        q_dots[:, i] = dynamics(ti, sol.y[:, i], alpha_fcn, params_)

    # Extract final position
    x_sol = sol.y[0, :]
    y_sol = sol.y[1, :]

    d = np.sqrt(x_sol[-1]**2 + y_sol[-1]**2)
    distances.append(d)
# Plot distance vs amplitude
plt.figure(figsize=(8, 6))
plt.plot(A_vals, distances)
plt.xlabel('Amplitude A')
plt.ylabel('Distance Traveled')
plt.title('Distance Traveled vs Gait Amplitude')
plt.grid(True)
plt.show()
```



Problem 2

Part 1

Write a Python function that takes a body twist vector and returns the body wrench from sliding snake-skin friction.

In [58]:
```
def snake_body_wrench(Vb, f_perp, f_par):
    epsilon = 1E-6
    # Vb is a body twist with wx, wy, wz, vx, vy, vz in body frame
    Fb = np.array([[0],
                   [0],
                   [0],
                   [0],
                   [0],
                   [0]])

    # Check if sliding forward/backward
    if np.abs(Vb[3]) < epsilon:
        Fb[3] = 0
    elif Vb[3] > 0:
        Fb[3] = -f_par
    else:
        Fb[3] = 2*f_par

    if np.abs(Vb[4]) < epsilon:
        Fb[4] = 0
    else:
        Fb[4] = -f_perp*np.sign(Vb[4])

    return Fb
```

#def snake_body_wrench_smooth(Vb, f_perp, f_par): # epsilon = 10 # Vb is a body twist with wx, wy, wz, vx, vy, vz in body frame # Fb = np.array([[0], [0], [0], [0], [0], [0]]) # Fb[3] = -f_par*(1.5*np.tanh(epsilon*Vb[3]+.3)+.3) # Checking if sliding forward/backward # Fb[4] = -f_perp*np.tanh(epsilon*Vb[4]) # return Fb

In [103…
```
def snake_body_wrench_smooth(Vb, f_perp, f_par):
    epsilon = 10

    Vb = np.asarray(Vb, dtype=float).reshape(6, 1)

    Fb = np.zeros((6, 1))

    # Smooth parallel friction
    Fb[3, 0] = -f_par * (1.5 * np.tanh(epsilon * Vb[3, 0] + 0.3) + 0.3)

    # Smooth perpendicular friction
    Fb[4, 0] = -f_perp * np.tanh(epsilon * Vb[4, 0])

    return Fb
```

Part 2

Write a function that takes in the configuration variables of the three-link swimmer and returns the net force from snake-skin sliding friction.

In [104…

```
# Part 2

from sympy import lambdify

q = sp.Matrix([x, y, theta, alpha1, alpha2])
qd = q.diff(t)

params = [l, k, A1, A2, omega, phi]

# Lambdify body twists
v_b_lambda = lambdify([q, qd, params], Vb_b, "numpy")
v_1_lambda = lambdify([q, qd, params], Vb_1, "numpy")
v_2_lambda = lambdify([q, qd, params], Vb_2, "numpy")

# Lambdify body Jacobians
J_b_lambda = lambdify([q, qd, params], Jb_b, "numpy")
J_1_lambda = lambdify([q, qd, params], Jb_1, "numpy")
J_2_lambda = lambdify([q, qd, params], Jb_2, "numpy")


def net_force(q_val, qd_val, params_val, f_par, f_perp):
    """
    q_val = [x, y, theta, alpha1, alpha2]
    qd_val = [xdot, ydot, thetadot, alpha1dot, alpha2dot]
    params_val = [L1, Lb, L2]
    """

    Vb = v_b_lambda(q_val, qd_val, params_val)
    V1 = v_1_lambda(q_val, qd_val, params_val)
    V2 = v_2_lambda(q_val, qd_val, params_val)

    Jb = J_b_lambda(q_val, qd_val, params_val)
    J1 = J_1_lambda(q_val, qd_val, params_val)
    J2 = J_2_lambda(q_val, qd_val, params_val)

    # Convert to numpy arrays
    Vb = np.asarray(Vb, dtype=float).reshape(6, 1)
    V1 = np.asarray(V1, dtype=float).reshape(6, 1)
    V2 = np.asarray(V2, dtype=float).reshape(6, 1)

    Jb = np.asarray(Jb, dtype=float)
    J1 = np.asarray(J1, dtype=float)
    J2 = np.asarray(J2, dtype=float)

    # Important: snake_body_wrench_smooth expects (Vb, f_perp, f_par)
    Fb = snake_body_wrench_smooth(Vb, f_perp, f_par)
    F1 = snake_body_wrench_smooth(V1, f_perp, f_par)
    F2 = snake_body_wrench_smooth(V2, f_perp, f_par)
```

```
    F = Jb.T @ Fb + J1.T @ F1 + J2.T @ F2

    return np.asarray(F, dtype=float).reshape(5)
```

Part 3

In [105…

```
from scipy.optimize import fsolve
import numpy as np

# Numeric parameters
l_val = 1.0
k_val = 1.0

A1_val = np.pi / 4
A2_val = np.pi / 4
omega_val = 2 * np.pi
phi_val = np.pi / 2

params_val = [l_val, k_val, A1_val, A2_val, omega_val, phi_val]

f_par = 1.0
f_perp = 2.0

# Pick one configuration and one shape velocity
q_val = np.array([
    0.0,   # x
    0.0,   # y
    0.0,   # theta
    0.0,   # alpha1
    0.0   # alpha2
])

alpha1dot_val = 1.0
alpha2dot_val = -1.0

def solve_body_velocity(V_guess):
    """
    Unknowns:
        V_guess = [xdot, ydot, thetadot]
    """

    xdot_val, ydot_val, thetadot_val = V_guess

    qd_val = np.array([
        xdot_val,
        ydot_val,
        thetadot_val,
        alpha1dot_val,
        alpha2dot_val
    ])

    F = net_force(q_val, qd_val, params_val, f_par, f_perp)
```

```
    # Force balance for floating body variables x, y, theta
    return F[0:3]

V_solution = fsolve(
    solve_body_velocity,
    x0=np.array([0.0, 0.0, 0.0])
)

print("Solved body velocity [xdot, ydot, thetadot]:")
print(V_solution)
```

```
Solved body velocity [xdot, ydot, thetadot]:
[-1.77335046e-02  0.00000000e+00  8.30623913e+02]
```

```
C:\Users\17076\AppData\Local\Temp\ipykernel_41668\1002490552.py:51: RuntimeWarning:
The iteration is not making good progress, as measured by the
  improvement from the last ten iterations.
  V_solution = fsolve(
```

Part 4

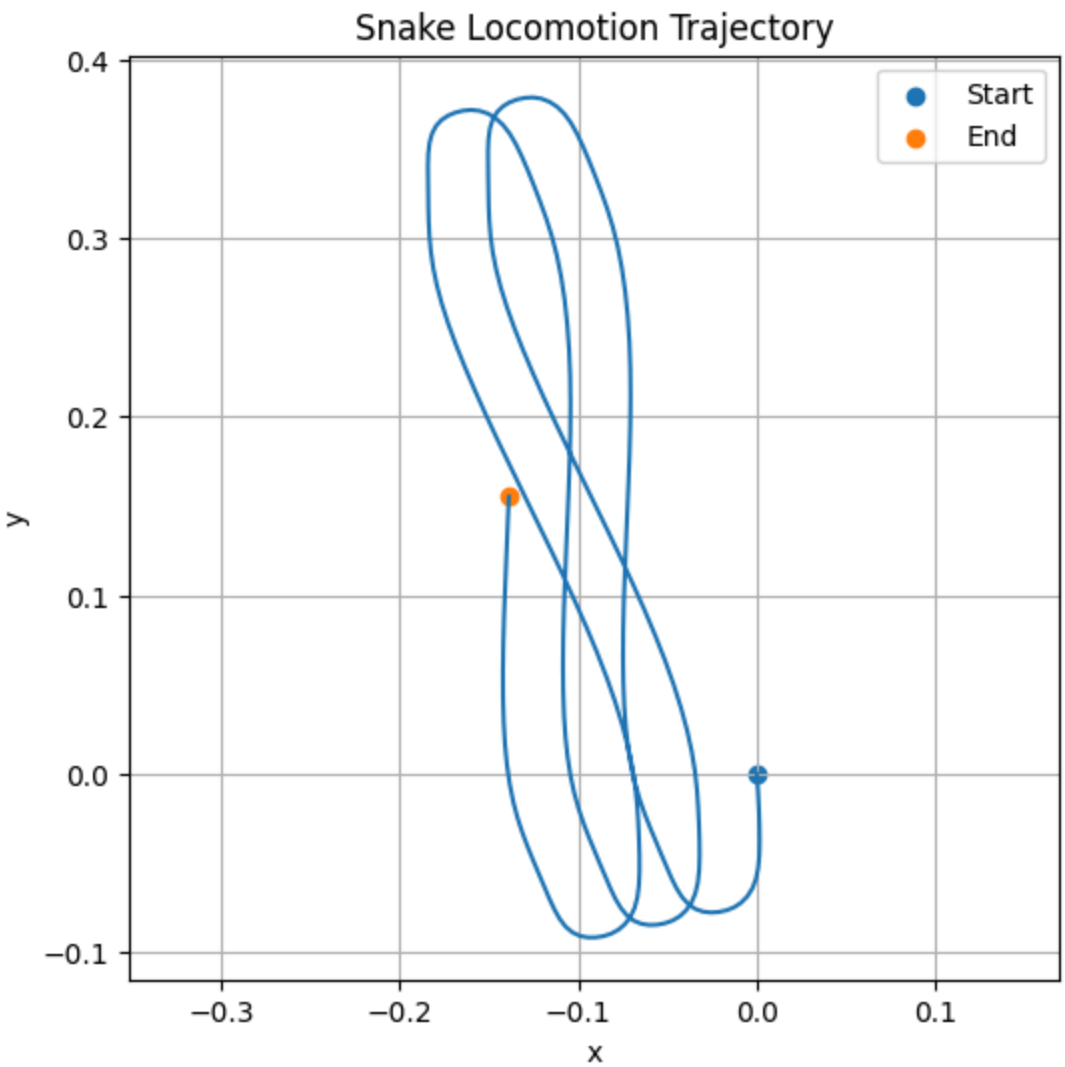
In [108…

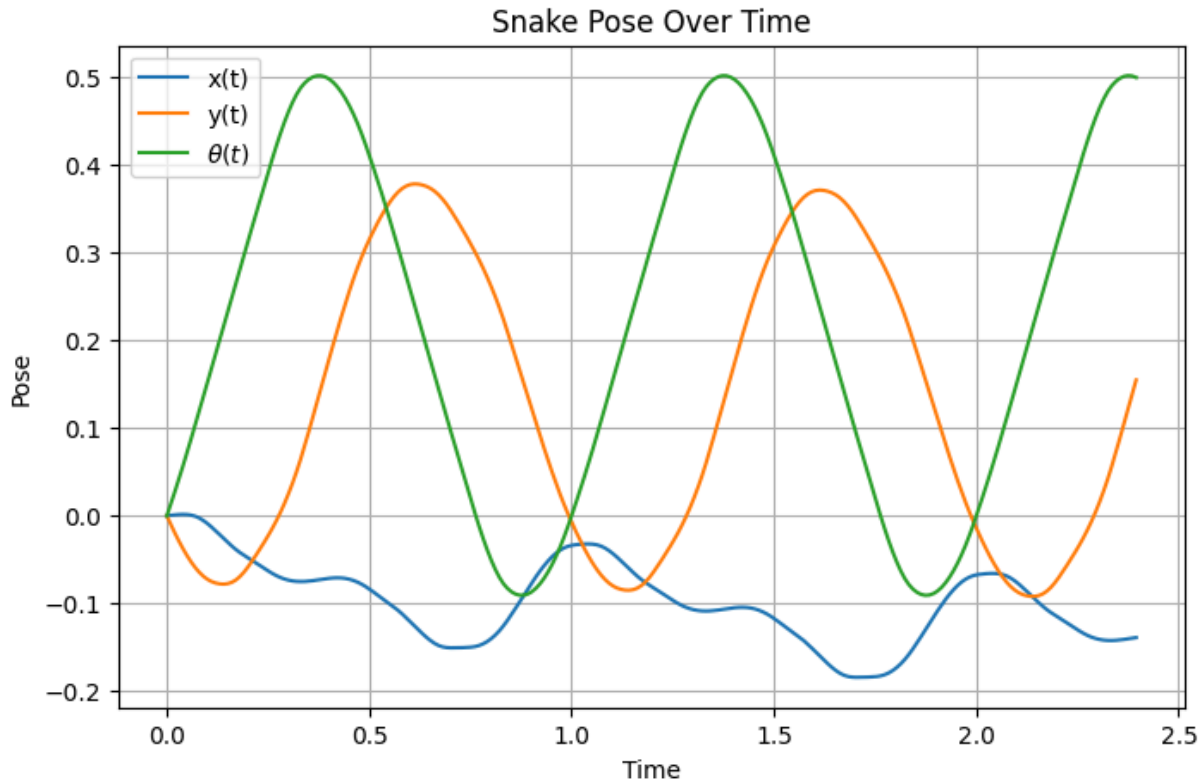
```
# Part 4 dynamics without global variable

from scipy.integrate import solve_ivp
from scipy.optimize import fsolve
import numpy as np
import matplotlib.pyplot as plt

# Store previous fsolve solution inside a list
V_prev = [np.array([0.0, 0.0, 0.0])]

def snake_dynamics(t_num, z):
    x_val, y_val, theta_val = z

    alpha1, alpha2, alpha1dot, alpha2dot = gait(t_num, params_val)

    q_val = np.array([
        x_val,
        y_val,
        theta_val,
        alpha1,
        alpha2
    ], dtype=float)

    def force_balance(V_guess):
        xdot, ydot, thetadot = V_guess

        qd_val = np.array([
            xdot,
            ydot,
            thetadot,
            alpha1dot,
            alpha2dot
        ], dtype=float)

        F = net_force(q_val, qd_val, params_val, f_par, f_perp)
```

```
        return F[0:3]

    # Use the previous solution as the next initial guess
    V_solution = fsolve(force_balance, V_prev[0])

    # Store solution for next timestep
    V_prev[0] = V_solution

    return V_solution

# Initial condition
z0 = np.array([
    0.0,   # x
    0.0,   # y
    0.0    # theta
], dtype=float)

# Time span
num_cycles = 5

omega_val = params_val[4]
period = 2 * np.pi / omega_val

t_span = (0.0, num_cycles * period)
t_eval = np.linspace(t_span[0], t_span[1], 500)

# Run simulation
V_prev[0] = np.array([0.0, 0.0, 0.0])

sol = solve_ivp(
    snake_dynamics,
    t_span,
    z0,
    t_eval=t_eval,
    rtol=1e-6,
    atol=1e-8
)

# Extract solution
x_sol = sol.y[0, :]
y_sol = sol.y[1, :]
theta_sol = sol.y[2, :]

# Plot trajectory
plt.figure(figsize=(6, 6))
plt.plot(x_sol, y_sol)
plt.scatter(x_sol[0], y_sol[0], label="Start")
plt.scatter(x_sol[-1], y_sol[-1], label="End")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Snake Locomotion Trajectory")
plt.axis("equal")
plt.grid(True)
plt.legend()
plt.show()
```

```
# Plot pose over time
plt.figure(figsize=(8, 5))
plt.plot(sol.t, x_sol, label="x(t)")
plt.plot(sol.t, y_sol, label="y(t)")
plt.plot(sol.t, theta_sol, label=r"$\theta(t)$")
plt.xlabel("Time")
plt.ylabel("Pose")
plt.title("Snake Pose Over Time")
plt.grid(True)
plt.legend()
plt.show()
```

```
C:\Users\17076\AppData\Local\Temp\ipykernel_41668\1675699030.py:40: RuntimeWarning:
The iteration is not making good progress, as measured by the
  improvement from the last ten iterations.
  V_solution = fsolve(force_balance, V_prev[0])
```




Snake Pose Over Time
x(t)
y(t)
θ(t)
Pose
Time
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
0.0
0.5
1.0
1.5
2.0
2.5